Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/08/2007
PI Name:	Ray, Chester A. Ph.D.		
Project Title:	Ultrasonic Bone Stimulation: Countermeasure to Orthostatic Intolerance		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	17033-2390	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	05/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-CA00404		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Orthostatic intolerance (OI) is a major physiological consequence of spaceflight. It has been estimated that nearly one-half to two-thirds of all astronauts experience OI after spaceflight. Post-spaceflight OI is a serious medical issue for manned-spaceflight because physical performance of an astronaut is severely impaired at a time when rapid egress from the spacecraft may be needed during an emergency. A number of physiological factors have been suggested to contribute to this problem with a number of countermeasures developed to prevent or alleviate post-spaceflight OI. However, as of yet, no single countermeasure has effectively prevented post-spaceflight OI. The overall objective of this project is to determine whether ultrasonic bone stimulation of the mastoid improves orthostatic tolerance and thus could serve as an effective and simple countermeasure for post-spaceflight OI. Based on preliminary data, it is hypothesized that ultrasonic bone stimulation of the mastoid will improve orthostatic tolerance in OI patients and in subjects exposed to bed rest. A secondary goal of this project is to determine the mechanism by which ultrasonic bone stimulation of the		

	mastoid increases orthostatic tolerance. It is hypothesized that this novel countermeasure acts to increase sympathetic nerve activity through engagement of the vestibulosympathetic reflex. Direct measurement of muscle sympathetic nerve activity will be made during ultrasonic bone stimulation in young, aged, and vestibular deficient subjects. This project should provide clear information on the possible role of ultrasonic bone stimulation of the mastoid as an effective and simple countermeasure for OI following spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current project may assist in our understanding of the mechanisms responsible for orthostatic intolerance and/or orthostatic hypotension in humans on earth. Additionally, the device being tested to elicit ultrasonic bone stimulation may be develop to help individuals on earth who experience orthostatic intolerance. This device is noninvasive and pose miminal risk to the patient.
Task Progress:	Our project in cooperation of the NASA bed rest study at UTMB-Galveston started examining the effect of ultrasonic bone stimulation on orthostatic tolerance in subjects exposed to prolonged head-down bed rest. Additionally, we examined how prolonged bed rest influences sympathetic nerve responses to vestibular activation. Four subjects were tested over the 90 day bed rest trial. All subjects showed decreased muscle sympathetic nerve responses (~75%) to head-down rotation (i.e., vestibular activation) by day 35 of bed rest and this attentuated response persisted at day 75 of bed rest. Thus, these early findings indicate an alteration of the vestibulosympathetic reflex to prolonged bed rest. We believe this may also be occurring with spaceflight and is one overlooked mechanism responsible for orthostatic intolerance after space flight. To date, we have only complete data on the effect of ultrasonic bone stimulation in one subject before and after bed rest. In this subject, ultrasonic bone stimulation improved orthostatic tolerance to a 80 degree head-up tilt test after bed rest. In 3 subjects we saw improved orthostatic tolerance with ultrasonic bone stimulation during pretesting. Thus, our one subject to date supports the concept that ultrasonic bone stimulation improves orthostatic tolerance.
Bibliography Type:	Description: (Last Updated: 01/26/2007)
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